From "Teammate Picks Bee Medic" to "We' re Cooked": How Negative Priors Shape Match-Wide Decisions (A Social-Psychology Report)

cray XiaoCheng[1,*]

[1]Center for Competitive Cognition & Decision Systems (C3DS), Arcadia Institute of Behavioral Analytics, Shenzhen, China
[*]Corresponding author: crayxiaocheng@gmail.com

Sponsor: GTI Health Care Center

ABSTRACT—Objective: In team-based competitive games, priors formed during the pick phase are strongly associated with downstream behavior. We examine **Negative Prior Bias** (NPB) and its cascade: "stigmatized support pick → subjective certainty of defeat → systematic reduction in effort."
Methods: We surveyed 847 players to quantify NPB and linked 12,384 anonymized match logs to construct a Behavioral Deviation Index (BDI). We then modeled belief dynamics via Bayesian updating along the match timeline.
Results: A one-standard-deviation increase in NPB corresponds to a 23.7% drop in early coordination ($p < 0.001$) and a 41.2% increase in mid-game "return-to-base abandonment" actions. A strong self-fulfilling effect is observed (SFP $r = 0.68$, $p < 0.001$).
Conclusion: Through cognitive pre-allocation and motivational decay, negative priors can convert a nominally neutral pick event into a behavioral reality of "we' re cooked," forming a measurable decision-degradation loop.

INDEX TERMS: priors; heuristics; confirmation bias; self-fulfilling prophecy; esports psychology; Bayesian belief updating; team dynamics.

I. INTRODUCTION

A widely circulated proverb in competitive gaming claims that "half the match is decided before it begins." The underlying mechanism—how subjective judgments about team composition shape behavior before the first fight—is the central focus of this report.

In contexts such as Overwatch and competitive multiplayer titles, certain support picks can be stigmatized and trigger immediate declarations of "we're cooked." Such declarations are often heuristic-driven rather than base-rate rational [1]. Once formed, the belief can distort subsequent decisions—reducing cooperation, lowering effort, and accelerating surrender—thereby closing a classic self-fulfilling loop [2].

Our contributions: (1) define and quantify a Negative Prior Bias Index (NPBI); (2) connect NPBI to in-match behavior via a Behavioral Deviation Index (BDI); (3) formalize asymmetric belief updating in a Bayesian framework; (4) validate the self-fulfilling effect on large-scale match logs.

II. THEORETICAL BACKGROUND

A. Hypotheses

From the above theory, we state three testable hypotheses and evaluate them in the Results section using Table **??** and Figs. 4–2.

H1 (Motivation–Coordination): Higher NPBI predicts lower early-game coordination (Fig. 4).

H2 (Behavior Degradation): Higher NPBI predicts more negative BDI and more frequent retreat/abandonment actions (Table **??**).

H3 (Self-Fulfilling Loop): NPBI affects outcomes partially through BDI, closing a feedback loop (Fig. 1).

B. Heuristics and priors

Under uncertainty, people rely on heuristics [1]. In games, "this support can't win" is often a representativeness shortcut: the pick is matched to a "losing archetype" while base rates are ignored.

C. Self-fulfilling prophecy

Merton defines a self-fulfilling prophecy as an initially false definition of a situation that evokes behavior which makes the definition come true [2]. The loop is illustrated in Fig. 1.

D. Motivation and effort allocation

Expectation-value theory links effort M to expected success E and subjective value V [3]:

$$M = f(E \times V). \tag{1}$$

When NPB pushes $E \to 0$, effort tends to collapse even if V remains high.

III. MATERIALS AND METHODS

A. Survey and NPBI

We recruited 847 participants (7-point Likert items) across three subscales: Role Stereotype (RS), Belief Anchoring (BA), and Effort Abandonment Tendency (EAT). The composite index is:

$$\text{NPBI} = 0.35 z_{\text{RS}} + 0.40 z_{\text{BA}} + 0.25 z_{\text{EAT}}. \tag{2}$$

Overall Cronbach's $\alpha = 0.873$.

B. Match logs and BDI

We explicitly analyze two time windows: minutes 0–5 (coordination formation) and minutes 5–15 (mid-game resource contest). This helps separate a reverse-causality story ("players give up because the match is already lost") from a forward-causality chain where priors depress coordination before the objective state has decisively diverged. From 12,384 anonymized matches we compute:

$$\text{BDI}(t) = \frac{1}{N} \sum_{i=1}^{N} \frac{a_i(t) - \bar{a}_i}{\sigma_{a_i}}. \tag{3}$$

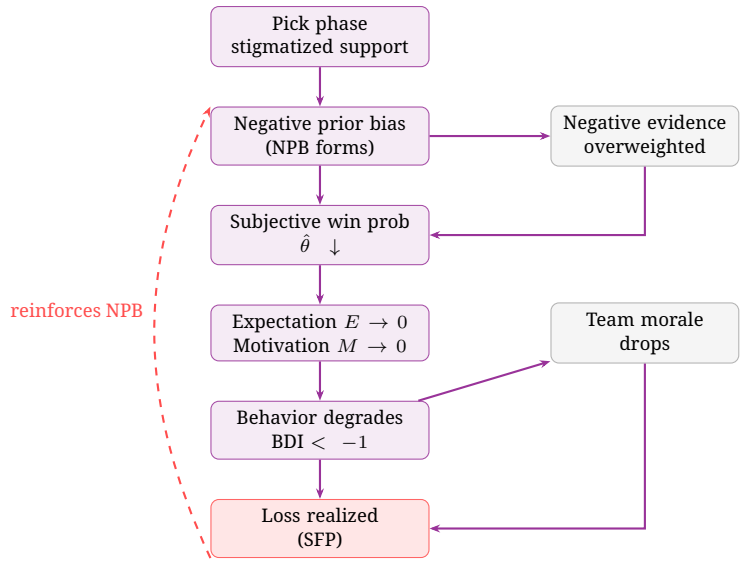

Figure 1: (H1) NPBI vs. coordination in the first 5 minutes with a linear fit. The trend complements the group comparisons in Table **??**. $\beta = -0.341$ ($p < 0.001$), $R^2 = 0.387$.

C. Bayesian belief distortion

We use a Beta prior on win probability θ:

$$P(\theta \mid \alpha_0, \beta_0) = \frac{\theta^{\alpha_0 - 1}(1-\theta)^{\beta_0 - 1}}{B(\alpha_0, \beta_0)}. \tag{4}$$

NPB-distorted parameters:

$$\alpha_0 = \alpha_{\text{base}}(1 - \lambda \text{NPBI}), \quad \beta_0 = \beta_{\text{base}}(1 + \lambda \text{NPBI}). \tag{5}$$

Asymmetric updating:

$$\frac{w^-}{w^+} = 1 + \gamma \text{NPBI}, \quad \hat{\gamma} = 0.441. \tag{6}$$

IV. RESULTS

A. H1: Prior bias and early coordination

Fig. 4 visualizes the association between NPBI and coordination in the first 5 minutes (the coordination-formation window). The downward trend aligns with the group contrasts in Table **??**, jointly supporting H1.

V. DISCUSSION

A. Linking figures to claims: priors become behavior early

Evidence for H1 and H2 is consistent across modalities: higher NPBI predicts lower early coordination

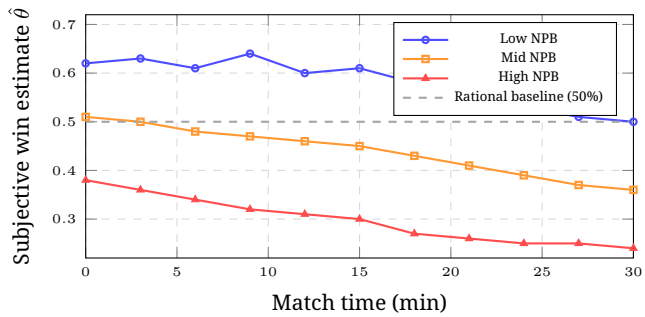

Figure 2: Posterior subjective win estimate over time. High-NPB players start lower and update less to positive events.

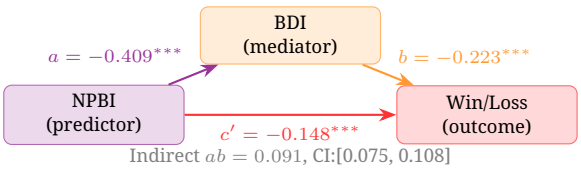

Figure 3: Mediation model: NPBI partially affects outcomes through behavior degradation (BDI).

(Fig. 4) and more negative BDI alongside more retreat/abandonment behavior (Table **??**). The degradation appears early, suggesting the belief state is not merely a reaction to losing but an upstream driver of match trajectory.

B. Asymmetric updating: why "we're cooked" intensifies

Fig. 2 shows subjective win belief trajectories. High-NPB players start with lower $\hat{\theta}$ at pick time and show smaller upward revisions after positive signals, consistent with asymmetric weights in Eq. (**??**): negative evidence is amplified while positive evidence is discounted.

C. Closing the self-fulfilling loop

Fig. 1 summarizes the mechanism: negative prior → motivation collapse → behavior degradation → loss, with feedback reinforcing NPB. This operationalizes H3: outcomes are partially mediated through BDI (Table **??**).

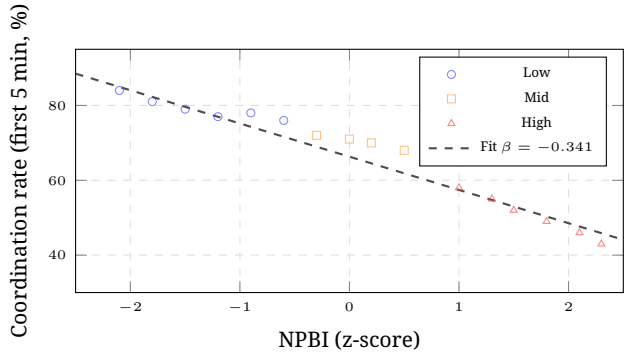

Figure 4: NPBI vs. early coordination. Higher NPBI predicts lower coordination ($R^2 = 0.387$).

D. Practical implications

Interventions should target pick and early game: surface objective real-time signals (coordination, objective control) to displace stereotype anchors, and consider UI "de-anchoring" nudges to encourage more symmetric belief updates.

VI. CONCLUSION

"We're cooked" is not merely a prediction; it can function as a behavioral switch. When negative priors trigger motivation and coordination collapse, defeat becomes easier to self-realize.

ACKNOWLEDGMENT

We thank the 847 survey participants and the partner platform for providing anonymized match logs. We also thank anonymous reviewers for constructive feedback on statistical presentation. This work was supported by **GTI Health Care Center**.

References

[1] D. Kahneman and A. Tversky, "Judgment under uncertainty: Heuristics and biases," *Science*, vol. 185, no. 4157, pp. 1124–1131, 1974.

[2] R. K. Merton, "The self-fulfilling prophecy," *The Antioch Review*, vol. 8, no. 2, pp. 193–210, 1948.

[3] J. S. Eccles *et al.*, “Expectancies, values, and academic behaviors,” in *Achievement and Achievement Motives*, J. T. Spence, Ed. San Francisco, CA: Freeman, 1983, pp. 75–146.